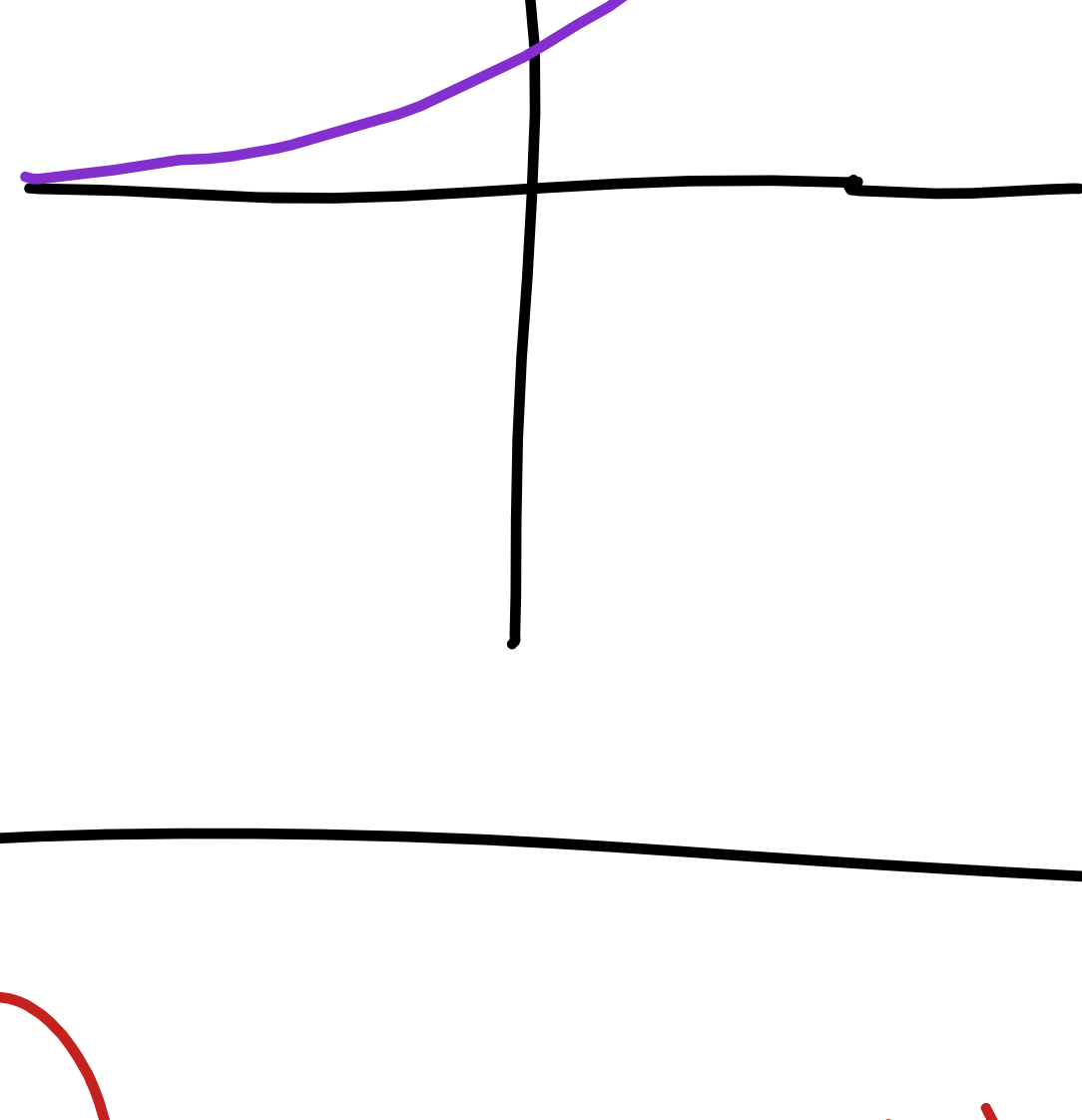


Deriv. and Anti-Deriv. of e

section 54

11/13/25

$$y = e^x$$



Derivative Rules!!! of e

$$\frac{d}{dx}[e^u] = e^u \cdot u'$$

$$\frac{d}{dx}[a^u] = a^u \cdot \ln(a) \cdot u'$$

$$\frac{d}{dx}[e^{3x^2}] = e^{3x^2} \cdot 6x$$

$$\begin{aligned} \frac{d}{dx}[x^3 e^{-2x}] &= \\ &= 3x^2(e^{-2x}) + x^3(-2e^{-2x}) \\ &= x^2 e^{-2x} (3 - 2x) \\ &= \frac{x^2(3-2x)}{e^{2x}} \end{aligned}$$

$$\frac{d}{dx}[\ln(5 + e^{4x})] =$$

$$\frac{4e^{4x}}{5 + e^{4x}}$$

$$\frac{d}{dx}\left[\frac{e^x + e^{-x}}{e^x - e^{-x}}\right] = \frac{d}{dx}\left[\frac{e^{2x} + 1}{e^{2x} - 1}\right] =$$

$$\frac{d}{dx}[x^3 + xe^y - y^5 = 2] =$$

$$3x^2 + xe^y \cdot \frac{dy}{dx} - 5y^4 \frac{dy}{dx} = 0$$

$$\frac{dy}{dx}(xe^y - 5y^4) = -3x^2$$

$$\frac{dy}{dx} = \frac{-3x^2}{xe^y - 5y^4}$$

Anti-Derivative

RULES!!!

of e

$$\frac{d}{dx}[e^u] = e^u \cdot u'$$

If we reverse this rule...

$$\int e^u du = \frac{e^u}{u'}$$

$$\int e^{3x+1} dx = \frac{1}{3} \int e^u du = \frac{1}{3} e^u + C$$

$$u = 3x+1$$

$$du = 3dx$$

$$\frac{1}{3} e^{3x+1} + C$$

$$\int 5xe^{-x^2} dx = -\frac{5}{2} e^{-x^2} + C$$

$$u = -x^2$$

$$du = -2x dx$$

$$\int \frac{e^{1/x}}{x^2} dx = -\int e^u du = -e^u + C$$

$$u = x^{-1}$$

$$du = -x^{-2} dx$$

$$\int \frac{e^{3x}}{(e^{3x} + 5)^2} dx = \frac{1}{3} \int u^{-2} du =$$

$$u = e^{3x} + 5 = \frac{1}{3} \cdot -u^{-1} + C$$

$$du = 3e^{3x} = -\frac{1}{3u} + C$$

$$= -\frac{1}{3(e^{3x} + 5)} + C$$

$$\begin{aligned} \int \frac{e^{7x} + e^x}{e^{3x}} dx &= \int (e^{4x} + e^{-2x}) dx \\ &= \frac{1}{4} e^{4x} - \frac{1}{2} e^{-2x} + C \end{aligned}$$

$$\int_{-1}^0 e^x \cos(e^x) dx = \int_{\frac{1}{e}}^1 \cos u du = \sin(u) \Big|_{\frac{1}{e}}^1$$

$$u = e^x$$

$$du = e^x dx$$

$$= \sin(1) - \sin\left(\frac{1}{e}\right)$$